Lab 1 - Revision - Subnetting, Routing & ACLs

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
HQ	G0/0	172.16.127.254	255.255.192.0	N/A
	G0/1	172.16.63.254	255.255.192.0	N/A
	S0/0/0	192.168.0.1	255.255.255.252	N/A
	S0/0/1	64.104.34.2	255.255.255.252	64.104.34.1
Branch	G0/0			N/A
	G0/1			N/A
	S0/0/0	192.168.0.2	255.255.255.252	N/A
HQ1	NIC	172.16.64.1	255.255.192.0	172.16.127.254
HQ2	NIC	172.16.0.2	255.255.192.0	172.16.63.254
HQServer.pka	NIC	172.16.0.1	255.255.192.0	172.16.63.254
B1	NIC			
B2	NIC	172.16.128.2	255.255.240.0	172.16.143.254
BranchServer.pka	NIC	172.16.128.1	255.255.240.0	172.16.143.254

Background / Scenario

This lab is intended to help refresh students on IP addressing, routing and access control lists. In this lab activity, you will finish the addressing scheme, configure routing and implement named ACLs.
Note: This lab is adapted from a different NetAcad one and also uses variables. As such, please do not attempt to copy work from any online sources or from any other students because, as well as that being plagiarism, it will not work anyway.

Step 1:Create Two Subnets For Use on Branch Router

Dividing a network into smaller subnets makes it easier for an administrator to implement security policies such as which subnets are allowed or not allowed to communicate together. It also reduces the size of our broadcast domain and so improves network performance. Devices are generally grouped into subnets based on factors like location (e.g. floors of building), departments or device types.

1. Divide 172.16.128.0/19 into two equal subnets for use on Branch.

QUESTION 1: What is the dotted decimal equivalent of the /19 subnet mask?

The subnet mask is made up of two portions, the network portion, and the host portion. This is represented in the binary by the ones and the zeros in the subnet mask.

QUESTION 2: What do the ones represent in the subnet mask?

QUESTION 3: What do the zeros represent in the subnet mask?

To subnet a network, bits from the host portion of the original network mask are borrowed in order to make subnet bits. The number of subnet bits defines the number of subnets.

QUESTION 4: How many bits need to be borrowed from the host portion in order to create two subnets? ***Hint****: The number of possible subnets created is calculated as 2^n where n equals the number of host bits borrowed.*

QUESTION 5: What would be the new subnet mask for the created subnets using the "slash notation" (e.g. /23)?

QUESTION 6: What would be the new subnet mask in dotted decimal notation (e.g. 255.255.254.0)?

QUESTION 7: What would be the network address of the first new subnet?

QUESTION 8: What would be the network address of the second new subnet?

Before proceeding, go to the "Lab 1 - Revision - Subnetting, Routing & ACLs - QUESTIONS" quiz on the Brightspace page and enter your answer for questions 1-8. Leave the quiz open while you complete the rest of the lab sheet.

a) Assign the first usable address of the second subnet to the Gigabit Ethernet 0/0 interface.

b) Assign the last usable address of the first subnet to the Gigabit Ethernet 0/1 interface.

c) Document the addressing in the Addressing Table.

d) Configure **Branch** with appropriate addressing.

2. Configure **B1** with appropriate addressing using the last available address of the network to which it is attached. Document the addressing in the Addressing Table.

Step 2:Configure OSPFv2

1. Configure **HQ** and **Branch** with OSPF routing according to the following criteria:

   a) Use process ID **10** for OSPF activation on both routers.

   Use the following command:

   ```
   HQ(config)# router ospf process-id
   ```

   b) Advertise all three attached networks. Do not advertise the link to the Internet. The **network statements** and **wildcard masks** are required to activate OSPF routing for the attached networks. The network statement values should be the network or subnet addresses of the configured networks.

   ```
   HQ(config-router)# network network-address wildcard-mask area area-id
   ```

   c) Configure appropriate interfaces as passive.

   ```
   HQ(config-router)# passive-interface interface
   ```

   QUESTION 9: Why do we configure certain interfaces as passive?

   Before proceeding, return to the quiz on the Brightspace page and enter your answer for question 9. Leave the quiz open while you complete the rest of the lab sheet.

2. Set a default route on **HQ** which directs traffic to S0/0/1 interface. Redistribute the route to **Branch**.

Step 3:Configure Standard Named Access Control Lists

1. From global configuration mode on **HQ**, enter the following command followed by a question mark.

   ```
   HQ(config)# ip access-list ?
   ```

   QUESTION 10 What are the two different type of ACLs that we can create?

2. To create a named standard ACL, enter the **standard** keyword, followed by an ACL name and press enter. You are now in named ACL configuration mode. Enter a question mark and press enter

   ```
   HQ(config)# ip access-list standard HQServer
   HQ(config-std-nacl))# ?
   ```

   QUESTION 11 Which of the available options would allow matched traffic to be forwarded?

   QUESTION 12 Which of the available options would allow matched traffic to be rejected?

3. To block traffic, enter **deny,** followed by a question mark.

   ```
   HQ(config-std-nacl))# deny ?
   ```

   QUESTION 13 What are the three types of sources that can be used to block traffic?

   Before proceeding, return to the quiz on the Brightspace page and enter your answers for questions 10-13.

4. Design a named access list **HQServer** to prevent any computers attached to the Gigabit Ethernet 0/0 interface of the **Branch** router from accessing **HQServer.pka**. All other traffic is permitted. Configure the access list on the appropriate router, apply it to the appropriate interface and in the appropriate direction.
   Note: For standard access control lists, the best practice is to place them as close as possible to the destination.

   QUESTION 14 Based on this requirement, what is the *source* of the traffic to be blocked by the HQServer ACL?

QUESTION 15 Since standard ACLs should be placed *as close as possible to the destination*, where should the HQServer ACL be applied and in which direction?

Before proceeding, return to the quiz on the Brightspace page and enter your answers for questions 14-15.

5. Design a named access list **BranchServer** to prevent any computers attached to the Gigabit Ethernet 0/0 interface of the **HQ** router from accessing the **Branch** server. All other traffic is permitted. Configure the access list on the appropriate router, apply it to the appropriate interface and in the appropriate direction.

If you have correctly configured all parts of the lab your activity score should now be showing as 100%. If so, click on "check results" in the activity window. Return to the Brightspace quiz one last time and enter the code into the appropriate question box (Q16) of the quiz.

You have completed the lab - please submit the Brightspace quiz.